Who benefits most?

Sorted partial effects · One clinical story, from model to inference

Valentin Klotzbücher · Department of Clinical Research

Our route

1. Literature and software.
2. One fictional heart-failure trial.
3. Sorted effects and patient classification.
4. Inference and clinical decisions.

Where this comes from

01

Describe the distribution of effects, then reconnect its tails to people.

↓ Two papers and one central idea

Two papers. One useful idea.

The method: Chernozhukov, Fernández-Val and Luo (2018), *Econometrica* (Chernozhukov, Fernández-Val, and Luo 2018).

The applied R tutorial: Chen et al. (2020), *The R Journal* (Chen et al. 2020).

Average effect

What is the overall benefit?

Sorted effects

How is benefit distributed?

Classification

Who occupies the tails?

The package already includes inference

`spe()`

Sorted effects and the average, with **bootstrap confidence bands**.

`ca()`

Tail characteristics, with **bootstrap standard errors and joint comparisons**.

We will use the original **SortedEffects 1.7.0** implementation for this trial.

The exact source version and numerical checks are documented. The package's inference still inherits its model, design and regularity assumptions.

One clinical question

02

After a heart-failure admission, who benefits from an intensified discharge strategy?

↓ A fictional randomized trial

Our fictional discharge trial

- **5,000 patients**, randomized at discharge after heart-failure admission.
- **Intensified diuretic strategy** versus the usual discharge regimen.
- Outcome: **unplanned readmission within 90 days**.
- Baseline age, a standardized **congestion index**, and a simulated female indicator.

Could a favorable average conceal patients with little benefit—or harm?

Entirely simulated. The treatment patterns and coefficients are invented for teaching; these are not clinical study findings.

Two profiles from the same fitted model

Both are age 60, with the same simulated sex indicator.

Lower congestion

10.4% → 11.7% readmission risk
Usual → intensified strategy

Higher congestion

34.9% → 18.6% readmission risk
Usual → intensified strategy

How would you summarize the benefit?

Different absolute benefits

Benefit = usual-care risk − intensified-strategy risk.

Lower congestion	Higher congestion
-1.3 pp 10.4% → 11.7%	16.4 pp 34.9% → 18.6%

Positive means fewer readmissions. Negative means more.

A benefit for each baseline profile

$$\Delta(x) = \mathsf{Pr}(Y = 1 \mid A = 0, X = x) - \mathsf{Pr}(Y = 1 \mid A = 1, X = x).$$

- Y: readmission within 90 days; $A = 1$: intensified strategy.
- The contrast is a **conditional average absolute benefit**.
- We choose the outcome, follow-up time and population **before sorting**.

Fit, predict, subtract, sort

03

A few lines of R connect the trial model to the sorted-effects curve.

↓ Follow the same 5,000 profiles

R · Fit the outcome model

```r
fm <- outcome ~ strategy * severity + age_z + female

fit <- glm(fm, data = trial, family = binomial())
```

The interaction lets the treatment odds ratio vary with baseline congestion.

`severity` = congestion index; `age_z` = age in decades, centered at 60. `outcome` = 90-day readmission.

R · Predict both strategies

```r
usual <- intensive <- trial
usual$strategy <- "usual"
intensive$strategy <- "intensified"

benefit <- predict(fit, usual, type = "response") -
  predict(fit, intensive, type = "response")
```

Keep each profile's baseline characteristics fixed. Change the treatment strategy.

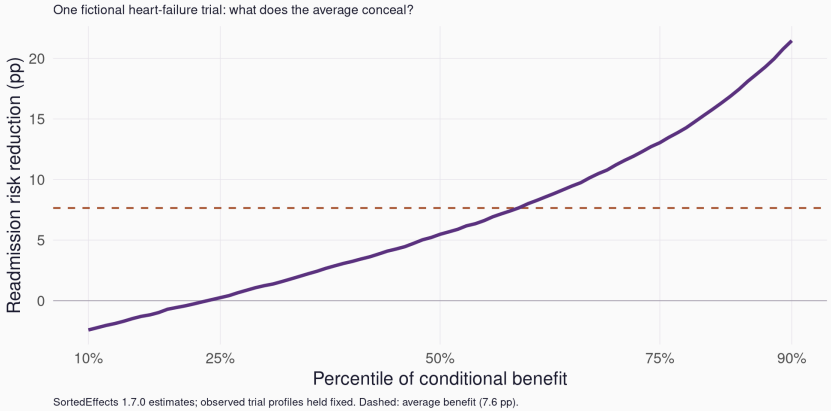
R · Summarize the distribution

```r
u <- seq(.10, .90, by = .01)
spe_curve <- Hmisc::wtd.quantile(
  benefit, weights = rep(1, nrow(trial)), probs = u
)
mean(benefit)
```

Mean: one overall benefit. **SPE:** benefit at each percentile.

The weighted-quantile convention matches the original package. The plotted range is the central 10th–90th percentiles.

The trial’s sorted benefit curve



One fictional trial. Estimated average benefit: 7.6 pp. Values below zero mean increased readmission risk under intensification.

Read a percentile carefully

The **80th percentile** is the benefit level at or below which approximately 80% of the profiles lie.

It is a **quantile of conditional benefits**.

It differs from a treatment effect on the 80th percentile of the outcome, or a probability that an individual patient will benefit.

Reconnect the tails to people

04

The curve shows how much. Classification asks who.

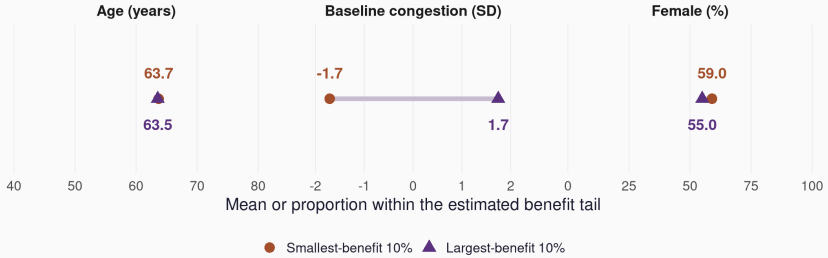
↓ Compare the lower and upper benefit tails

R · Describe the tails

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(trial)), c(.10, .90)
)
groups <- list(low = benefit <= cutoff[1],
               high = benefit >= cutoff[2])
lapply(groups, function(i)
  colMeans(trial[i, c("age", "severity", "female")]))
```

This calculation describes the groups. It does not yet give their uncertainty.

Classification · Read the dumbbells



Original ca() tail means. Separate axes retain each characteristic's units. Connecting lines are not confidence intervals; inference for differences follows.

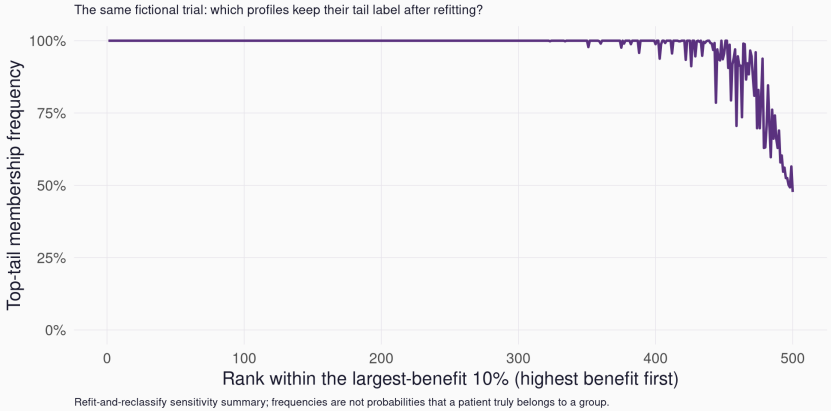
Each line connects the mean characteristic in the lowest- and highest-benefit 10%. Axes retain their own clinical units; connecting lines are not confidence intervals.

What the profile differences mean

- **Direction:** which characteristics are more common or larger in the highest-benefit group?
- **Size:** is the difference clinically interpretable in its original units?
- **Uncertainty:** how stable is the comparison after refitting and reclassification?

A profile difference describes the tails of this fitted model. It does not establish an individual treatment rule.

Tail membership can change after refitting



Refit-and-reclassify sensitivity diagnostic for this trial. Frequencies are not probabilities that an individual truly belongs to a benefit group.

Put uncertainty around the story

05

The package refits and re-sorts. Classification also needs reclassification.

↓ Confidence bands and joint comparisons

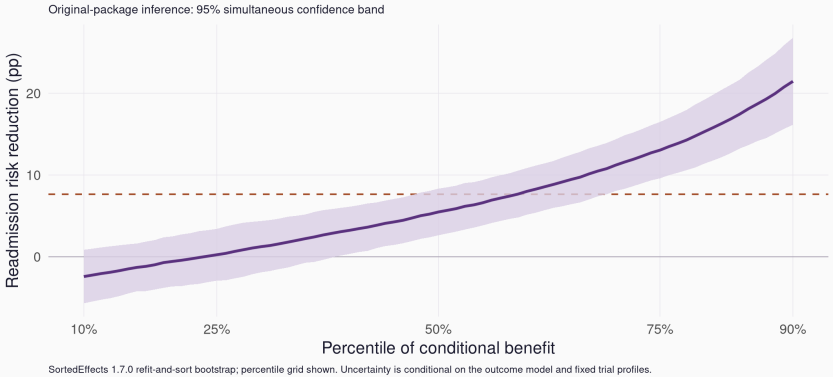
R · SPE inference is implemented

```r
spe_fit <- SortedEffects::spe(
  fm = fm, data = trial, method = "logit",
  var_type = "categorical", var = "strategy",
  compare = c("intensified", "usual"),
  us = u,
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

Output: estimated SPEs, average effect, bootstrap standard errors and uniform confidence bounds.

Here `compare` is ordered to obtain usual minus intensified risk; its sign was checked against direct predictions.

The SPE curve with its confidence band



Original SortedEffects 1.7.0: 95% uniform bootstrap band over the displayed percentile grid, 499 repetitions, uncorrected point estimate.

R · Classification inference is implemented

```r
ca_fit <- SortedEffects::ca(
  fm = fm, data = trial, method = "logit",
  var_type = "categorical", var = "strategy",
  compare = c("intensified", "usual"),
  u = .10, interest = "moment", cl = "diff",
  t = c("age", "severity", "female"),
  b = 499, alpha = .05, bc = FALSE, seed = 420
)
```

Question tested: do mean characteristics differ between the highest- and lowest-benefit groups?

Which profile differences are convincing?

Characteristic	High minus low	Bootstrap SE	Joint p
Age (years)	-0.15	2.87	1.00
Congestion (SD units)	3.45	0.06	<0.01
Female (percentage points)	-4.00	4.00	0.63

Joint p-values account for comparing these three characteristics.

Congestion separates these modeled tails; age and sex show little evidence of a difference.

Original `ca()` moment comparisons; high-benefit minus low-benefit tail. Rounded Monte Carlo results from 499 repetitions. These are not tests of treatment benefit within a subgroup.

What inference still depends on

The design: randomization and a defined population.

The model: credible conditional risks and prespecified effect modifiers.

The procedure: appropriate resampling, tail sizes and joint comparisons.

Sampling uncertainty is one part of the evidence. Model sensitivity and clinical validation remain essential.

Back to the treatment decision

06

Would this evidence justify intensifying treatment for a particular profile?

↓ Discuss the same clinical question

Would you change the discharge regimen?

For the fictional highest-benefit group, ask:

- Is the estimated reduction in readmission large enough relative to **harms and burden**?
- How uncertain are the contrast and the **tail label**?
- Does the model work in a **new clinical setting**?
- Does a proposed treatment rule perform well in **independent evaluation**?

Three things to take away

Define the benefit.

Show its distribution and tails.

Use inference—and challenge the model.

Open discussion

1. What does this add to our usual interaction or subgroup analyses?
2. What would you challenge first in this clinical model?
3. What evidence would justify using the ranking in practice?

Use the chapter map to return to the SPE curve, dumbbells or inference table.

Reading and resources

Method: (Chernozhukov, Fernández-Val, and Luo 2018).

R tutorial: (Chen et al. 2020).

The session page contains the executable walkthrough, simulation, results and software notes.

Thank you — where would this be useful in our own work?

A closer look

+

Optional definitions, a scale illustration, and software validation.

↓ Technical backup

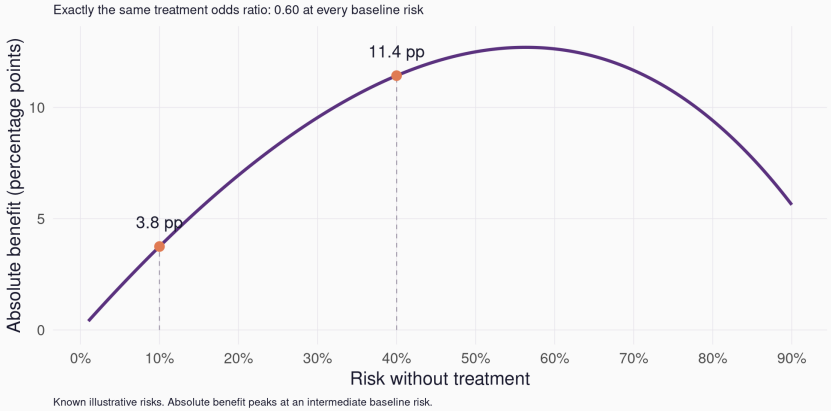
Backup · What is being sorted?

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)].$$

- The distribution of X defines the population being summarized.
- Classification reconnects the tails to baseline characteristics.
- A quantile of effects differs from a difference of outcome quantiles.

(Chernozhukov, Fernández-Val, and Luo 2018). Main estimates use the original package's weighted empirical quantile convention.

Backup · The scale matters



Optional analytic illustration: a constant odds ratio of 0.60 still gives varying absolute risk differences. It is not the main trial’s interaction model.

Backup · Simulation assumptions

- Baseline logit includes age, congestion and the simulated sex indicator.
- The treatment log odds ratio varies with congestion.
- The fitted specification matches the simulated data-generating model.
- The same 5,000 observed trial profiles define the primary SPE population.

R/demo.R contains coefficients, seeds and diagnostics. A constant-OR comparison is retained for software exercises, outside the main story.

Backup · Read the resampling target

`spe()` **full-sample path:** refits with bootstrap weights, then takes quantiles using the original profile weights.

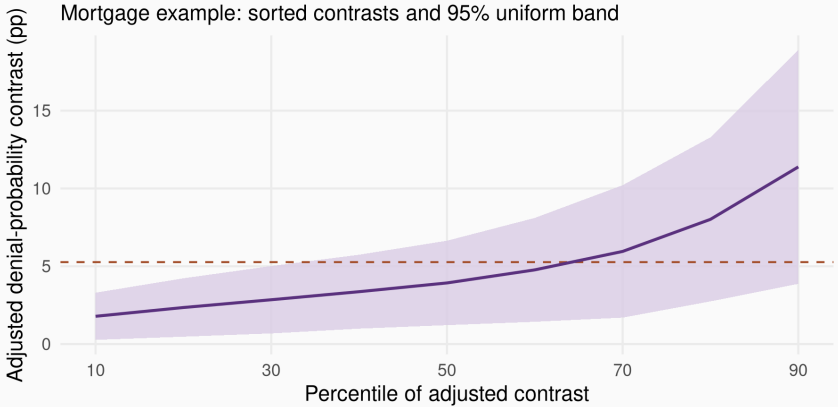
`ca()` **moment path:** refits, updates tail thresholds and recomputes characteristic means using resampled weights.

Membership diagnostic: a separate refit-and-reclassify sensitivity calculation.

Backup · What the package test reports

- `ca(..., interest = "moment", cl = "diff")` compares high- minus low-benefit mean characteristics.
- Joint p-values adjust across the **specified characteristics**.
- Bootstrap SEs and joint p-values are reported directly.
- Simultaneous confidence intervals for moment differences are **not inferred from those p-values**.

Backup · Original published example



Original tutorial's mortgage data (Munnell et al. 1996; Chen et al. 2020). Separate software benchmark, not an additional clinical case in the main presentation.

References

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, "SortedEffects: Sorted causal effects in R," *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, "The sorted effects method: Discovering heterogeneous effects beyond their averages," *Econometrica*, 86 (2018), 1911–1938.

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, "Mortgage lending in Boston: Interpreting HMDA data," *The American Economic Review*, 86 (1996), 25–53.